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Solar System Missions: Clementine

Mapping the Moon

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Introduction

Clementine is the lunar probe that quietly reopened the Moon for the United States. Launched on 25 January 1994, more than twenty-one years after Apollo 17's astronauts left the lunar surface, Clementine was the first American craft to circle the Moon since the Apollo era, and it did more than revisit old ground: it mapped the entire Moon in eleven spectral bands, measured the global topography with a laser altimeter, and produced the first evidence, from its bistatic radar experiment, that water ice might hide in the permanently shadowed craters of the lunar south pole, a claim that redirected two decades of lunar exploration and was vindicated, for real water, by the missions that followed.

The mission's story is unusual in the space age's roster: Clementine was not primarily a science mission at all. It was a technology demonstration for the Pentagon's missile-defense bureau, the agency descended from the Strategic Defense Initiative, whose engineers wanted to test lightweight sensors and spacecraft components against the hard radiation and thermal swings of deep spaceflight; the Moon happened to be the ideal test range, and a small team at the Naval Research Laboratory, flying cheap and fast, built a spacecraft that happened to return some of the most scientifically productive lunar data of the century.

The chapters ahead follow the full arc: the mission's origins in missile-defense engineering and its bargain with NASA; the spacecraft's design, instruments, and flight through phasing loops to lunar orbit; the two months of mapping that gave planetary science its first global mineralogical and topographic models of the Moon; the radar experiment and the ice controversy; the failed attempt to fly by the asteroid Geographos; and the mission's legacy, from the prospector missions it inspired to the modern lunar programs that build on its maps. The mission took its name from the American folk song about the miner's daughter who is lost and gone forever, an in-joke from engineers who expected their cheap spacecraft to be exactly that; instead, this book tells how it was found again, in the data it left behind.

A note on the mission's record before the story begins: Clementine flew for less than five months, mapped the Moon for two of them, and has been cited in the literature for thirty years; this book's account rests on the mission's published results, the ice paper of 1996, the global maps and topography released through the planetary data archives, and the post-mortem analysis of the Geographos failure, all of them public documents the reader can consult. The chapters keep the record's precision where it matters, the dates, the depths, and the radar's caveat, because the mission's story is one of numbers honestly earned and honestly argued, and the honest argument, as much as the data, is what this book has to teach.

The discovery chapter should hold the project's double origin honestly, the partnership that built it: a defense department office with instruments and a rocket to spare, a civilian agency with science to do and no budget for it, the mission, proposed as a joint venture, the Moon, mapped as a by-product of a technology demonstration. The reader who finds the hybrid strange can find it honest, the craft, a scout for both masters, the partnership, uneasy and productive, the origin, the mission's defining fact from the first memo.

Chapter 1: The Discovery of Clementine

The discovery story of Clementine is a story of the peace dividend of missile-defense engineering, and it begins in the Pentagon rather than at a space agency.

The Ballistic Missile Defense Organization, heir to the Strategic Defense Initiative's sensor programs, had spent a decade developing lightweight components for interceptor spacecraft: compact cameras, radiation-hardened computers, star trackers, and inertial measurement units, all built small and fast for weapons that never flew. The end of the Cold War left the technology needing a flight test and the agency needing a mission, and the solution was ingenious: use the Moon and an asteroid as a deep-space test range, exercising every component against the radiation, the thermal swings, and the navigation demands of real interplanetary flight, while a NASA science team rode along to use the sensors for lunar exploration. The arrangement, formalized in the early 1990s, gave the mission its dual name, the Deep Space Program Science Experiment, and its unconventional customer structure: the Pentagon paid for and flew the spacecraft, and NASA funded the science team, a bargain that produced a lunar mission for roughly eighty million dollars, an order of magnitude cheaper than the flagship probes of the era.

The spacecraft was built at the Naval Research Laboratory by a small, fast team in the spirit the era called faster, better, cheaper: twenty-two months from start to launch, a three-hundred-kilogram machine crammed with miniaturized instruments, launched on 25 January 1994 from Vandenberg atop a Titan II, itself a converted Cold

War intercontinental missile, the whole mission an exercise in turning swords into plowshares with the paperwork barely changed. The mission's name, taken from the folk song about the miner's darling daughter, was the team's own joke about the spacecraft they expected to be lost and gone forever, and the name stuck as the mission's official call sign.

The flight plan matched the name's gallows humor with ambition: five days of Earth phasing loops to line up the lunar geometry, lunar orbit insertion on 19 February 1994, two months of systematic mapping from a low polar orbit, and then, if the spacecraft survived, a gravity-assisted departure for an August flyby of the asteroid 1620 Geographos, a close pass that would have returned the first resolved images of an Earth-approaching asteroid. The spacecraft survived the Moon, accomplished its mapping, and died trying: a software fault in May 1994, on the way out, left the spacecraft spinning with its attitude thrusters firing until the propellant was gone, and the asteroid flyby was lost, a failure mode the mission's engineers analyzed with a candor that itself became a textbook case.

The discovery of Clementine, in the sense this chapter means, is the discovery that the mission's cheap, militarized design worked: the sensors performed, the navigation held, and the science return, global multispectral maps, the first laser altimetry of the Moon, and the polar radar experiment, came from hardware built for another war. The rest of this book tells what the data said.

The discovery chapter should name the mission's people, because small missions are personal in a way flagships never are: the program's leadership at the Ballistic Missile Defense Office and the Naval Research Laboratory, the engineering teams who built the spacecraft in twenty-two months, and the NASA science team, led by the mission's own scientists, who negotiated the observing plan that turned a sensor test into a survey. The mission's personalities, a program manager famous for fast decisions and a science team that argued the mapping into existence, are documented in the era's reporting, and the reader who wants the mission's flavor can take it from the record's own detail, a spacecraft built by people who expected to lose it, planning to survey a world on the way.

The discovery chapter's close belongs to the launch itself, which the record preserves as the mission's perfect emblem: on 25 January 1994, a converted Titan II, a missile built to end the world, rose from Vandenberg carrying a spacecraft built to test the missile-defense systems of the world's next war, aimed by a team whose song choice joked about the mission's death, at the Moon, the peacetime destination of the whole species' first space age. The layers are almost too neat, and the historian's discipline is to report them without editorializing: the launch happened, the layers were real, and the mission that followed justified the emblem by reopening the Moon that the same rockets had once helped close.

One more discovery-chapter credit belongs to the mission's institutional ancestor, the Strategic Defense Initiative, whose sensor research of the 1980s, pursued for interceptor programs that never deployed, produced the component technologies Clementine flew, and the honest history notes the chain: the initiative's cameras, trackers, and computers, miniaturized for weapons that never flew, orbited the Moon in a test the weapons had funded. The chain is the peace dividend's lunar chapter, and the discovery story's full credit line runs from the initiative's labs through the conversion of the early 1990s to the small team that flew the result, a lineage the mission's participants cited themselves, with the wryness the era allowed.

The spacecraft's shape deserves a word of description, the octagonal bus that carried the survey: a squat drum, roughly a meter and a half across, the sensors, mounted on its faces, the attitude, held by four small thrusters and a spinning pair of momentum wheels, the craft, small enough to fit in a modest living room. The reader who pictures planetary probes as great machines can recalibrate, the bus, the size of a washing machine, the mission, flown on a modest chassis, the whole spacecraft, lighter than a small car and cheaper than most.

The understanding chapter should also hold the mission's chronology in one paragraph, the schedule as it ran: approved in 1992, launched in January 1994, mapping through spring, the failure in May, the asteroid leg cancelled, the spacecraft retired that summer, the science, running years past the mission, the announcement, two years later, the whole flight, over in under half a year. The reader who wants the mission's length can have it exactly, the active life, brief as a semester, the influence, thirty years and counting, the brevity, the mission's own trivia.